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19 / 20 submitted

Q1. Union of Sets

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}

print(A.union(B))
```

INPUT

```
A= {1,2,3}
B= {3,4,5}
```

OUTPUT

```
{1, 2, 3, 4, 5}
```

Q2. Intersection

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

result = A.intersection(B)
print(result)

result2 = A & B
print(result2)
```

OUTPUT

```
{2, 3}
{2, 3}
```

Q3. Difference

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

print(A - B)
```

OUTPUT

{1}

Q4. Symmetric Difference

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}

print(A ^ B)
```

OUTPUT

{1, 4}

Q5. Subset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {2, 3}
B = {1, 2, 3, 4}

if A.issubset(B):
    print("Subset")
else:
    print("Not Subset")
```

OUTPUT

Subset

Q6. Superset

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3, 4}
B = {2, 3}

if A.issuperset(B):
    print("Superset")
else:
    print("Not Superset")
```

OUTPUT

Superset

Q7. Disjoint Sets

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2}
B = {3, 4}

if A.isdisjoint(B):
    print("Disjoint")
```

```
print("Disjoint")  
else:  
    print("Not Disjoint")
```

OUTPUT

Disjoint

Q8. Remove Duplicates

Score: 0.00 TC: 0/1 passed

CODE

```
lst = [1, 2, 2, 3, 3, 4]  
s = set(lst)  
print(s)
```

INPUT

List = 1 2 2 3 3 4

OUTPUT

{1, 2, 3, 4}

Q9. Common Elements

Score: 5.00 TC: 1/1 passed

CODE

```
A = {1, 2, 3}
B = {2, 3, 4}
C = {2, 5}

common = A & B & C
print(common)
```

OUTPUT

{2}

Q10. Missing Numbers

Score: 0.00 TC: 0/1 passed

CODE

```
given = {1, 2, 4, 5, 7, 10}

all_numbers = set(range(1, 11))
missing_numbers = sorted(all_numbers - given)

print(set(missing_numbers))
```

INPUT

{1,2,4,5,7,10}

OUTPUT

{8, 9, 3, 6}

Q11. First Repeated

Score: 5.00 TC: 1/1 passed

CODE

```
arr = list(map(int, input().split()))

seen = set()
for x in arr:
    if x in seen:
        print(x)
        break
    seen.add(x)
```

INPUT

10 20 30 20 40

OUTPUT

20

Q12. Duplicate Elements

Score: 0.00 TC: 0/1 passed

CODE

```
numbers = [10, 20, 30, 20, 10]

seen = set()
duplicates = set()

for num in numbers:
    if num in seen:
        duplicates.add(num)
    else:
        seen.add(num)

print("Original List:", numbers)
print("Duplicate Elements:", list(duplicates))
```

INPUT

10 20 30 20 10

OUTPUT

Original List: [10, 20, 30, 20, 10]
Duplicate Elements: [10, 20]

Q13. Compare Lists

Score: 0.00 TC: 0/1 passed

CODE

```
def compare_unique_elements(list1, list2):
    return set(list1) == set(list2)

list1 = [1, 2, 2, 3]
list2 = [3, 2, 1]
print(compare_unique_elements(list1, list2))
```

INPUT

List1: 1 2 2 3
List2: 3 2 1

OUTPUT

True

Q14. Duplicate Tuples

Score: 0.00 TC: 0/1 passed

CODE

```
t = [(1,2),(1,2),(3,4)]  
result = list(set(t))  
print(result)
```

INPUT

```
[(1,2),(1,2),(3,4)]
```

OUTPUT

```
[(1, 2), (3, 4)]
```

Q15. Pair Sum

Score: 0.00 TC: 0/1 passed

CODE

```
def pair_sum(arr, target):
    pairs = set()
    n = len(arr)
    for i in range(n):
        for j in range(i + 1, n):
            if arr[i] + arr[j] == target:
                pairs.add((arr[i], arr[j]))
    return sorted(pairs)

line1 = input().strip()
nums = []
for part in line1.replace("List", "").replace(":", " ").replace("=", " ").split():
    nums.append(int(part))
arr = nums

line2 = input().strip()
target = int(line2.split("=")[1].strip()) if "=" in line2 else int(line2.split()[-1])

result = pair_sum(arr, target)
print(",".join(f"({a},{b})" for a, b in result))
```

INPUT

List : 2 4 6 8 10
Target=12

OUTPUT

(2,10),(4,8)

Q16. Unique Characters

Score: 5.00 TC: 1/1 passed

CODE

```
s = input().strip()
unique_chars = set(s)
print(len(unique_chars))
```

INPUT

Programming

OUTPUT

8

Q17. Unique Words

Score: 0.00 TC: 0/1 passed

CODE

```
sentence = input().strip()
words = set(sentence.split())
print(words)
```

INPUT

python is easy python

OUTPUT

{'python', 'is', 'easy'}

Q18. Exactly One Set

Score: 0.00 TC: 0/1 passed

CODE

```
A = {1, 2, 3}
B = {3, 4, 5}
C = {5, 6, 7}

result = (A - B - C) | (B - A - C) | (C - A - B)

print(result)
```

OUTPUT

{1, 2, 4, 6, 7}

Q19. Menu Driven**CODE**

```
def read_set(prompt):
    return set(map(int, input(prompt).split()))

A = read_set("Enter the first set elements: ")
B = read_set("Enter the second set elements: ")

while True:
    print("\nMenu")
    print("1. Union")
    print("2. Intersection")
    print("3. Difference (A - B)")
    print("4. Symmetric Difference")
    print("5. Exit")
    ch = input("Enter your choice: ").strip()

    if ch == '1': print("Union:", A | B)
    elif ch == '2': print("Intersection:", A & B)
    elif ch == '3': print("Difference:", A - B)
    elif ch == '4': print("Symmetric Difference:", A ^ B)
    elif ch == '5':
        print("Program terminated.")
        break
```

```
else:
    print("Invalid choice.")
```

INPUT

OUTPUT

```
Traceback (most recent call last):
  File "C:\Windows\TEMP\sandbox_192521404RD_6a71f752134658.11012989\main.py", line 14, in <module>
    ch = input("Enter your choice: ").strip()
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
EOFError: EOF when reading a line

mmetric Difference
5. Exit
Enter your choice:
```

— Not Submitted —